

## **6A<sup>th</sup> Grade: Unit 15, Lesson 6**

1. A game requires rolling a five on a fair six-sided die on the first try in order to win the game. After six hundred tries, 100 people roll a five. Is this a possible or an impossible result?
2. What happens to the probabilities when the number of trials in a simulation increases?
3. Alicia predicted that if she rolled a fair six-sided die 20 times, the die would land on an even number 10 times. The theoretical probability of rolling an even number is 50%. She then performed a simulation, and an even number was rolled 240 out of 400 total rolls. Is the probability of rolling an even number based on her simulation greater than or less than the theoretical probability of landing on an even number?